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Framing Migration in the Context of the Skilled Labor Shortage: A Stance Detection Analysis of German Newspaper Coverage

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Abstract

In the recent years tensions between an increasingly apparent economic necessity of foreign workers and an intensified societal refusal of immigration are observable, not only in Germany but worldwide. Now as ever, newspapers play an important role in shaping public discourse, being both agent and resonance chamber of public discourse. This paper will conduct the attempt of monitoring the stance of two major German newspapers on migration in the last two years, making use of current computational methods in stance detection. The goal is to gather insights into the frequency of reporting about migration and the shortage of skilled workers and whether the observed newspapers react to sociopolitical key events with an altered latent stance towards migration. Since automated stance detection of newspaper articles is a barely examined research field, this paper will explore the capabilities and limits of BERT-based and generative large language models in stance detection for a consumer-hardware-friendly and cost-effective environment, aiming to evaluate their practicality for scalable, real-world applications. Finally, I will conduct an exemplary analysis of the data, further exploring what insights can be gained from this type of research attempt.

1. Introduction

The shortage of skilled labor power in western industrial nations has become an urging and continuously growing challenge for the labor markets of these countries. Due to the demographic change, the successive retirement of the generations of the 1950s and 1960s with a historically high birthrate, the labor market in Germany is threatened to reach a point where the present and future need of labor force exceeds by far the supply, thus possibly leading to recession and consequently fueling social conflicts and insecurities. The labor force, that is the relation between the total population and the working population of a country, keeps getting smaller due to a low birthrate of the younger and the successive retirement of the "boomer" generation. One obvious way to address this problem is immigration, to increase the national labor force by integrating foreign workers to the labor market and keep the national economy at a stable level, while also taking into account the willingness of many foreign workers to move to Germany not only for economic, but also for safety reasons (Kuhn *et al.*, 2025, pp. 1–3).

Projections indicating that without immigration Germany's labor force would shrink by 15 percent by 2035 underscore the urgency of the issue (Fuchs, Söhnlein and Weber, 2021, p. 3). Other measures to increase labor force – such as raising the retirement age or raising the full-time employment rate among women – would still fall far short of meeting the necessary demand of

labor force. Besides, the natural limits of integrating more elderly, women and currently unemployed to the job market, are already nearly reached¹ (Fuchs, Söhnlein and Weber, 2021, p. 3). Beyond that, it is normatively questionable, if this would be a desirable imagination of a labor economy, since it would mean an increased need of day nursery (and consequently more qualified workers) and delayed retirement. Considering the already historical low rates of unemployment in Germany, as well as the surplus load of care work for the majority of women, this "resource" is already almost depleted. For a scenario without immigration, the predicted labor force by 2040 would reach roughly 41,13 million (Fuchs, Söhnlein and Weber, 2021, p. 7). A recent study assumes a minimal necessary labor force of at least 45,7 million workers by 2040 to keep the economy just about running, that is to keep the current standard of living, and consequently a necessary net immigration of at least 288.000 people per year, up until 368.000 (Kubis and Schneider, 2024, p. 63).

Parallel we witness an unprecedented agitation against immigrants and asylum seekers the recent years worldwide. In Germany, it is heavily driven by the right-winged party AfD and increasingly adopted by the conservative and liberal parties CDU/CSU and FDP. According to the professor for integration research and social policy at Humboldt University Naika Foroutan, an increasing feeling of stagnation and socioeconomic demise fell together with the great flight immigration in the years 2015/2016 as well as the migration wave as an aftermath of the war in Ukraine since 2022. This would lead to the impression of a de facto causal relationship of migration and a socioeconomic structural crisis, conveniently being reproduced and utilized by right-winged political actors to make out a scapegoat for the current crisis, while addressing racist residuals in large parts of society (Foroutan *et al.*, 2024, pp. 4–6). While an elaboration of the reasons for this phenomenon would by far exceed the limits of this paper, it is a clear fact, that the dilemma is growing between an increasing economic demand for immigration, especially in the sector of skilled workers, so called "Fachkräfte"², and a public discourse that is more and more shaped by the rejection of immigration, especially of asylum seekers. Surveys among labor migrants from non-EU countries in Germany show that 56% say to have experienced racism towards them in the recent year (Fendel and Ivanov, 2024, p. 6). While Germany is on the one hand one of the most popular countries for occupational migration, it is also among the ones where foreign workers stay the least and are the least happy, so

1 Other studies disagree with this exposition, highlighting that a raise of the retirement age would unlock great potentials of raising labor force (Kubis and Schneider, 2024, p. 13). What remains unclear, however, is how this approach would avoid only postponing the problem. Neither does it consider normative questions, nor possible rebound effects.

2 The Term "Fachkraft" typically lines up orderly in the following hierarchy: Helper, Fachkraft, Specialist, Expert; and has a prerequisite a completed two years of vocational training or respectively skills and special knowledge (Kuhn *et al.*, 2025, p. 2). While I try to stick to this definition I have to note, that in political communication, as well as in public discourse, the different terms are partly used synonymously, leading to ambiguities in its meaning.

that they often leave after a few years (InterNations, 2023). Almost one third of migrant workers leaving Germany don't leave because of better job offers elsewhere or for family reasons, but because of "other reasons". Surveys show that among those other reasons discrimination and racism, such as discontent due to a lack of social integration seem to be the major causes, especially among people from Northern Africa and the Middle East (Loschert, Leisenheimer and Komitowski, 2025, pp. 30–33).

Despite the growing influence of social media, traditional media outlets such as newspapers continue to play a significant role in shaping public opinion in Germany – even among their critics (Arlt and Wolling, 2018, p. 93). As mass media discourse strategies shape public opinion by influencing how social problems are perceived, newspaper coverage has a direct impact on the public perception of migration and consequently on political preferences and policies (Schetsche, 2001, p. 38). While many (post-)industrial nations compete increasingly for skilled workers worldwide, they are simultaneously confronted with a growing xenophobic discourse against migrants and asylum seekers, which may ultimately become an economic threat in itself (Diermeier *et al.*, 2025, pp. 14–16; Newbery, 2025, p. 7). Therefore, experts advocate for a de-escalation of the rhetoric about migration and the promotion of a positive outline – be it for reasons of a better social climate or economic considerations (Foroutan *et al.*, 2024, p. 9; Loschert, Leisenheimer and Komitowski, 2025, pp. 57–59).

Since the issue described tends to bear a significant potential for social conflict, it is of sociological interest to monitor the media discourse about migration and the shortage of skilled workers in Germany in order to better understand the role of newspapers within the production and reproduction of public discourse. A quantified research of media bias can therefore lead to valuable insights into how traditional media outlets reflect and cover the social climate. In this paper, I will therefore apply stance detection as a computational method to map the positions of two major German newspapers on migration within the discursive context of Germany's skilled labor shortage. Given that stance detection in newspaper articles has so far been rarely applied, this paper also aims to explore the potentials of this method for accessible, low-threshold research endeavors. The above mentioned considerations lead to the following research questions:

[RQ1] What are the capabilities and limitations of computational methods – specifically NLI classifiers and generative large language models – for stance analysis in newspaper articles?

[RQ2] How is migration framed in the context of "Fachkräftemangel", the shortage of skilled workers, in major German newspapers?

[RQ3] What stances do different newspapers take and how does stance distribution change over time?

Hypotheses for RQ2 and RQ3 are that key events do not only influence the frequency of discourse on migration, but also shape the stance taken by journalists and newspapers toward migration. This effect may even be visible within the discourse about migration and labor shortage, even when the events are not directly related to the specific issue. Another hypothesis is that a conservative newspaper will report significantly more negative about migration than the progressive one.

2. Literature review

Stance, in distinction to sentiment, is the expression of an opinion towards a certain target. Instead of determining exclusively the emotionality of language as sentiment does, it aims to identify the directionality of an expression towards a well-defined target, also named *hypothesis*. Therefore stance is much more valuable when it comes to analyzing political opinion in texts, since it includes context and meaning and therefore meets the requirements of real world natural language much better, reducing ambiguities that often occur in cases of irony or indirect speech with sentiment-detection (Burnham, 2024, pp. 2–3). Although political stance can take shape in various complex forms, to keep it simple, I will follow the example of the majority of researchers referenced, limiting it to the classifiers: supportive, neutral, and oppositional towards the target.

While the majority of stance-detection research focuses on the automatic annotation of short texts like X (Twitter) posts, headlines or individual sentences, the expansion of the research towards entire newspaper articles is a very recent development, bringing along a variety of complex problems (Darwish *et al.*, 2020; ALDayel and Magdy, 2021; Burnham, 2024; Mets *et al.*, 2024). Longer texts offer, on the one hand, more context to classify the stance of an article towards the target, on the other hand, they add complexity to the task of annotation, not only for the machine but also for human annotators. Traditional media, especially newspaper outlets, ideally follow journalistic standards of objectivity and a balanced coverage of the situation, including a variety of viewpoints in order to enable the readers to form an informed opinion (Arlt and Wolling, 2018, p. 76). When it comes to automatic annotation of newspaper articles by machine-learned algorithms, these journalistic standards, "objectivity" and "detachment", make it especially difficult to identify a clear directedness without knowledge of further context about the topic and discussion (Feree *et al.*, 2002, cited in Riedl, 2024, p. 42). The subtleties expressing political bias and stance in traditional

media, make it particularly challenging to automate their annotation and let algorithms assign a stance label with confidence. Moreover, the opinion bias in newspaper articles is often shaped by journalistic decision making such as "source selection", "narrative framing" and "lexical subtleties", that require substantial context and language understanding to interpret accurately (Lee *et al.*, 2025, p. 3). Recent research developments in the automatic stance detection of news articles with generative LLMs address this problem in several ways.

Mets *et al.* (2024), for example, work exclusively on a sentence level, splitting newspaper articles into individual sentences based on topic-relevant keywords and then mapping, manually annotating, and classifying each sentence, using LLMs based on BERT (Bidirectional encoder representations from transformers) to detect their stance towards immigration. This approach, as they argue, has the advantage of potentially identifying multiple stances within single documents, but comes at the expense of missing valuable contextual information (Mets *et al.*, 2024, p. 5).

Another short-segment stance detection approach is via so called NLI (Natural Language Inference) classifiers presented by Burnham (2024), which are language models that are pre-trained as universal classification tools, assigning the most likely (or the least unlikely) label to a hypothesis. These NLI classifiers are easy to use, efficient and have a good performance, but they lack context information, especially if the target or information is not mentioned explicitly in the provided text (e.g. names, idioms or slogans). They have stricter limitations to their application than generative models and may require different models for different stance topics (Burnham, 2024, p. 6). Burnham also promotes in-context classification based on generative LLMs for annotation tasks of short texts, though he highlights concerns around reproducibility, the high costs of proprietary models, and notes that open models do not perform significantly better than NLI classifiers (Burnham, 2024, p. 11).

Another BERT based classification approach is put to the test by Mascarell *et al.* (2021), who use a dataset that has stance annotations on an article level, supplemented by emotion annotations on article and paragraph level. However they face difficulties when it comes to the annotation of entire articles, since their LLM (in 2021) could process only up to 512 tokens, but their articles have an average length of 843 ranging up to over 4000 tokens (Mascarell *et al.*, 2021, p. 73). It can be assumed that contemporary models are more capable concerning this matter.

An even more sophisticated approach is attempted by Lee *et al.* (2025), who separate the articles in "journalism-guided segments" (Lee *et al.*, 2025, p. 2), such as headlines, leads, paragraphs and quotations and let a language model (LM) annotate each one of them separately. While longer texts have a significantly negative influence on the performance of LLMs, their approach makes usage of the good capabilities of LMs to annotate short text segments, delegating this information to the

prompt of a second generative LLM to make the final article-level classification, taking into account all the individual classifications from the first step (Lee *et al.*, 2025, pp. 5–6). This multilevel analysis over several LLMs requires obviously more computational and financial resources than the ones above.

Each approach has its up- and downsides and in the end it comes to pondering the weighs and losses when considering how to proceed with ones own research. In the following, I will combine some of the methods mentioned above and operate at the paragraph level, maintaining a reasonable token limit per request. To provide additional context to the language model, I will include title and lead of each article. Furthermore, I will evaluate the agreement scores of both an NLI classifier and an in-context classification approach using open-source generative LLMs, comparing them against a human-labeled subset of the data.

3. Data

I chose as a dataset a selection of newspaper articles from two opinion-forming German newspapers with different political orientations *Die Welt* and *Frankfurter Rundschau (FR)*, based on accessibility via LexisNexis and the fact that the newspapers represent opposing political orientations. The articles have been filtered by a combination of keywords, finally forming two corpora, one covering broadly the discourse about migration and work, explicitly excluding terms like "Fachkräfte" (dataset A), and the other specifically enclosing the discourse about migration and shortage of skilled workers (dataset B). For the exact keywords used in the LexisNexis query, see Appendix A 1.1. As a time frame I chose the period from 01. June 2023 until 15. June 2025, covering a time period of just over two years. Within this time frame, several topic relevant events took place, such as the stepwise implementation of a reform of the so called "Fachkräfteeinwanderungsgesetz" (FEG), a legal framework that has as a goal to counter the shortage of skilled workers by acknowledging foreign diplomas and vocational trainings and simplify the allocation of work visa (DIE ZEIT, 2023). Furthermore, the 2025 federal elections, along with the preceding election campaign that strongly focused on immigration, as well as several terrorist attacks in Mannheim, Solingen, and Munich, contributed to a heated-up and increasingly negative discourse around migration (Eikmanns and Sittenauer, 2025).

After loading the articles into a data frame, I proceeded with further cleaning of the data, separating title, text body, metadata, date and newspaper into individual columns. Following the removal of duplicates, I split the articles into paragraphs, filtering them based on relevance, using a regular expression pattern to identify keywords appearing in each paragraph. This was the most convenient

way to narrow the data down to topic-related paragraphs. The keywords for Dataset A contain general migration related keywords and work related keywords. For Dataset B, I used a combination of keywords related to the shortage of skilled workers, in conjunction with the migration-related keywords from Dataset A (see Appendix A.1.2). Additionally, interviews were automatically detected, and each question-answer pair was merged into a single paragraph whenever matching keywords were found in either part, in order to preserve contextual information. For better processing, I dropped paragraphs with more than 400 words, most of them cases that lacked paragraph separation, thus often including entire articles. After assigning an identifier to each article in each dataset and merging the data, I ended up with a total of 733 relevant articles, containing 868 relevant paragraphs. The distribution of paragraphs over the newspapers was very balanced with 50.23% being from *FR* and 49.77% being from *Die Welt*. In the final dataset for automatic processing, the average length of a paragraph was 103.66 words, which is equivalent to roughly 150-170 tokens (see Appendix A 1.3).

4. Methodology

As mentioned above, the goal is to automate the process of stance classification of different paragraphs. In order to test this, I used a human-labeled subset of my data of 250 paragraphs. In the following I conducted a performance test with three different NLI-Classification models and two generative LLMs. To optimize model performance, I tried several of the above mentioned methods and adjustments such as including more context information by adding titles and leads and different prompting techniques suggested by Burnham (2024, p. 12).

4.1 Annotations

For evaluation of the models performance, I took a random sample of 250 of the 868 paragraphs (~29%) to have a human-labeled reference to test the capabilities of the different models used. To each paragraph I added the title of the article to provide some additional context for annotation. I labeled the paragraphs with specific guidelines in *positiv*, *negativ* and *neutral/irrelevant* towards immigrants and immigration. The fundamental guideline for annotation was: "The text talks rather *positive/negative/neutral* about migrants or migration". Further followup questions were: "does the article reproduce anti-immigrant sentiment?" and "does the article see immigration rather as a problem or a solution to socioeconomic challenges?". Articles that were not about the subject of interest were classified as *neutral/irrelevant* (Appendix B). During the annotation process it became apparent that the dataset had been well selected, with the majority of paragraphs actually referring

to the subject of interest and with a relatively low share of irrelevant articles. I tried to link my annotations as close to the actual expression of the paragraphs, disregarding context information, especially about people, organizations and parties, that would let me assume the stance of someone, concentrating on the literal meaning of the paragraph. There were two main difficulties in annotating, that appeared during the process: First were paragraphs that talk about the difficulties that migrants have to face in everyday life. These paragraphs were often written in a negative tone, highlighting social issues and discrimination. Technically these could be classified as *negativ* since they talk about negative migration experiences and thus could also be interpreted as having a negative stance towards migration, but certainly not negative towards immigrants. I labeled these paragraphs as neutral, since the research interest focuses rather on resentment against migrants reproduced by the media. The second type of edge cases, although few in number, consisted of paragraphs that primarily oppose and criticize discrimination and racist public discourse about immigration. However, despite their critical stance, these paragraphs often unintentionally reproduced racist stereotypes by repeating them. Because they perpetuate harmful stereotypes – even if to criticize them – these paragraphs were labeled as negative. I noted some borderline cases for clarification of the annotation guidelines and improvement of LLM prompting (Appendix B).

4.2 NLI Classifiers

NLI classifiers are universal zero-shot classifiers that have been trained to detect "[...] if the meaning of one text (the hypothesis) is entailed in another text (the premise)." (Laurer *et al.*, 2024, p. 2). Given a document as premise, the classifier evaluates multiple hypotheses and returns a probability score of each of the labels: *entailment*, *contradiction* (clearly no entailment) or *neutral*. The final result matches the hypothesis with the highest probability of entailment. For example in stance-detection, three hypotheses represent each stance: "Der Text äußert sich {positiv/negativ/neutral} gegenüber Migrantinnen und Migranten.". The final label is selected as the one with the highest entailment score, according to the NLI classifier (Laurer *et al.*, 2024, pp. 2–3). For my annotation, I used three different NLI classifiers via a huggingface pipeline and a variety of different hypotheses prompts. Because different hypotheses can significantly influence the performance of NLI classifiers, I tried to optimize templates for each model used (see Appendix C) (Burnham, 2024, p. 10). I could not identify one specific template that consistently performed better across models, but rather each model had its corresponding optimized template. However some templates performed better across different models. Each model took as an input (premise) the title and paragraph, separated by line breaks. As suggested by Lee *et al.* (2025), I tried to improve

context knowledge by adding the lead paragraph, typically illustrating a short introduction to the article, after the title, but could only achieve slightly better results with one model. The test results are shown in Table 1, with optimized context and prompting techniques for each of the models. Computational time for 250 paragraphs was steadily under 2 minutes on a T4 GPU in Google Colab.

4.3 In-Context Classifiers

The second approach was to automatically annotate the dataset with generative LLMs and test their performance. In-context learning is a way to provide practical guiding to generative LLMs by forwarding task instructions into the input prompt (Lee *et al.*, 2025, p. 5). Since generative LLMs are multi purpose tools, they can easily adapt to new tasks, making them useful in a variety of natural language classification tasks. Although they require significantly more computational power than NLI classifiers, these models can easily be provided with additional contextual information, which may improve their classification capabilities, especially of socioeconomically complex topics such as immigration and the shortage of skilled workers. With a structured output JSON-schema, a generative LLM can be forced to answer in the JSON syntax which makes it easier to extract labels in the specific syntax from the answer. I compared two non-proprietary LLMs with different prompting and context provision. In general, computational time took significantly longer than with NLI classifiers. For 250 leads and paragraphs, both models took around 6 minutes for annotation on a T4 GPU in Google Colab.

5. Results & Discussion

The results show that both NLI and in-context classifiers do not perform exceptionally well but still achieve a reasonable level of agreement with the human-labeled data. Similar results with BERT based classifiers are achieved by Lüüsi *et al.* (2024, p. 19) annotating Estonian newspaper articles. With a maximum F1 score of only 0.578, the performance of the models is currently insufficient for reliable use on real-world data, highlighting the need for further improvement.

Table 1: Entailment classification metrics

NLI Model	Cohens Kappa	F1	Accuracy	Precision	Recall
MoritzLaurer	0.228	0.452	0.460	0.564	0.460
mlburnham	0.379	0.578	0.584	0.621	0.584
luissattelmayer	0.195	0.389	0.428	0.538	0.428

Table 2: In-context classification metrics

gLLM	Cohens Kappa	F1	Accuracy	Precision	Recall
gemma3:4b	0.310	0.536	0.532	0.602	0.532
llama3.1:8b	0.280	0.533	0.552	0.566	0.552

By tendency, the in-context classifiers achieved better results than most NLI classifiers but still fell short of the best-performing NLI classifier by Michael Burnham *et al.*:

- `mlburnham/Political_DEBATE_DeBERTa_large_v1.1` (Burnham *et al.*, 2024)

What is remarkable is, that this classifier is actually not trained on a German data corpus. However it seems to perform better than the other two multilingual models. Among the in-context classifiers, the `gemma3:4b` model reached similar results with the best performing NLI classifier, but obviously at the cost of significantly higher ecological and computational expenses. It is known that NLI classifiers perform particularly well, when the target (here: migration) is mentioned. Due to data filtering, the target (or related terms) are represented in every paragraph, which could be an explanation for the comparatively good performance of the NLI classifier (Burnham, 2024, p. 6).

Table 3: `mlburnham/Political_DEBATE_DeBERTa_large_v1.1` detailed evaluation metrics

	Precision	Recall	F1	Support
negativ	0.67	0.66	0.67	62
neutral	0.69	0.41	0.52	109
positiv	0.48	0.76	0.59	79
accuracy			0.58	250
macro avg	0.62	0.61	0.59	250
weighted avg	0.62	0.58	0.58	250

By taking a closer look at the evaluation metrics of the best performing NLI classifier, it becomes clear that the model has difficulties with the *neutral* class. A recall of 0.41 means, that the model only finds 41% of the total neutral paragraphs. A precision of 0.69 is not particularly impressive because the neutral class is overrepresented in the dataset. Therefore, the chances of being correct when predicting neutral are higher. Due to its ambiguity the neutral class however is the hardest to detect, especially in newspaper articles where a formal and objective tone is predominantly used. The linguistic subtleties that express a certain stance towards migrants and migration are difficult to detect, even for a human annotator, and often require background knowledge. The *negativ* class performed more reliable with a balanced precision and recall, leading to a decent F1 score of 0.67. That means, most of the negatives that were predicted were correct, while the model found at least

66% of all negatives. The *positiv* class was imbalanced with the model catching 76% of the positives, but often over-predicting actual non-positives as positives.

The results show that even the most advanced models used for this specific task quickly reach their limitations. The best model with an F1 macro score of 0.578 is acceptable but not particularly impressive. Other researchers doing stance detection of newspaper articles achieve better results with fine tuning their models, but this requires significantly more effort and is not an out-of-the-box use case (Mascarell *et al.*, 2021; España-Bonet, 2023; Burnham *et al.*, 2024; Cruickshank and Ng, 2024; Mets *et al.*, 2024; Lee *et al.*, 2025). When fine-tuning is not an option, NLI classifiers appear to be a valid alternative, given their performance and the efficiency – at least for getting an initial overview of the data.

Since the models did not perform well enough yet to be used in real-world scenarios, I narrowed the task down to a less complex binary classification problem, distinguishing into a binary classification of *negativ* and *nicht negativ* (non-negative). This required narrowing down the research question to focus specifically on negative reporting about migration and work, which I considered a valid and reasonable adjustment. By merging the the *positiv* and *neutral* class into *nicht negativ*, I was able to significantly improve the models performance even without finetuning, achieving the following results:

Table 4: Metrics of binary classification across models

Model	Cohens Kappa	F1	Accuracy	Precision	Recall
gemma3:4b	0.276	0.668	0.644	0.761	0.644
mlburnham	0.556	0.831	0.828	0.835	0.828

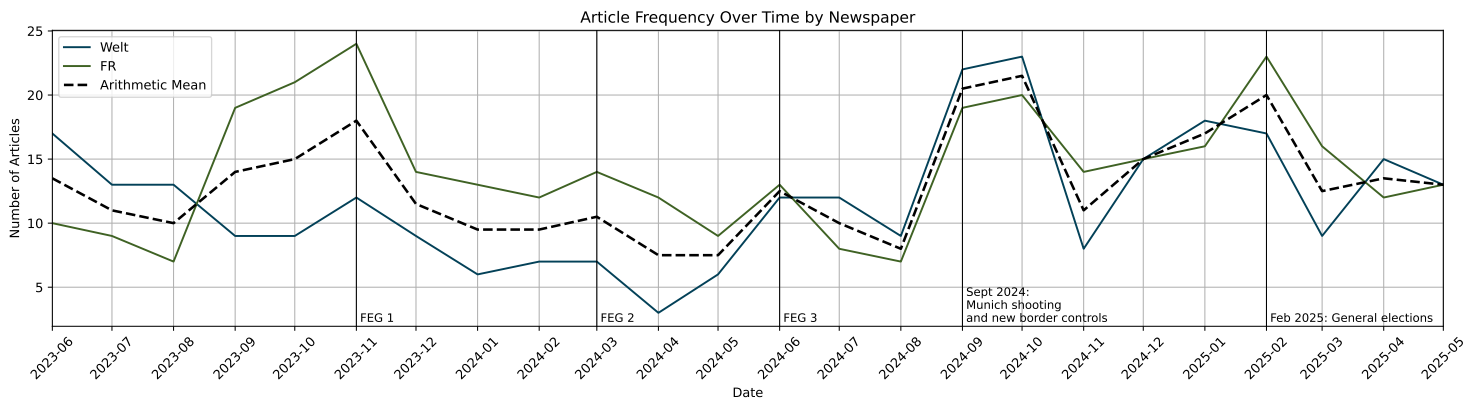
The significantly better performance on this simplified task enabled me to proceed with the following exemplary analysis of the dataset. I used the NLI classifier by Michael Burnham to annotate the remaining data, resulting in a fully labeled dataset with a promising potential of coming close to the quality of human annotation.

6. Exemplary analysis

Finally, I will conduct an exemplary analysis of the data to demonstrate the research capabilities of the above presented method and to find answers for the remaining research questions. To gain a better overview of the data, I plotted article frequency over the selected time period, with each data point representing the total number of articles published during that month. I marked some key

events in the plot that, as expected, resulted in an increased output of articles about the topic by both newspapers.

Figure 1: Article frequency over time

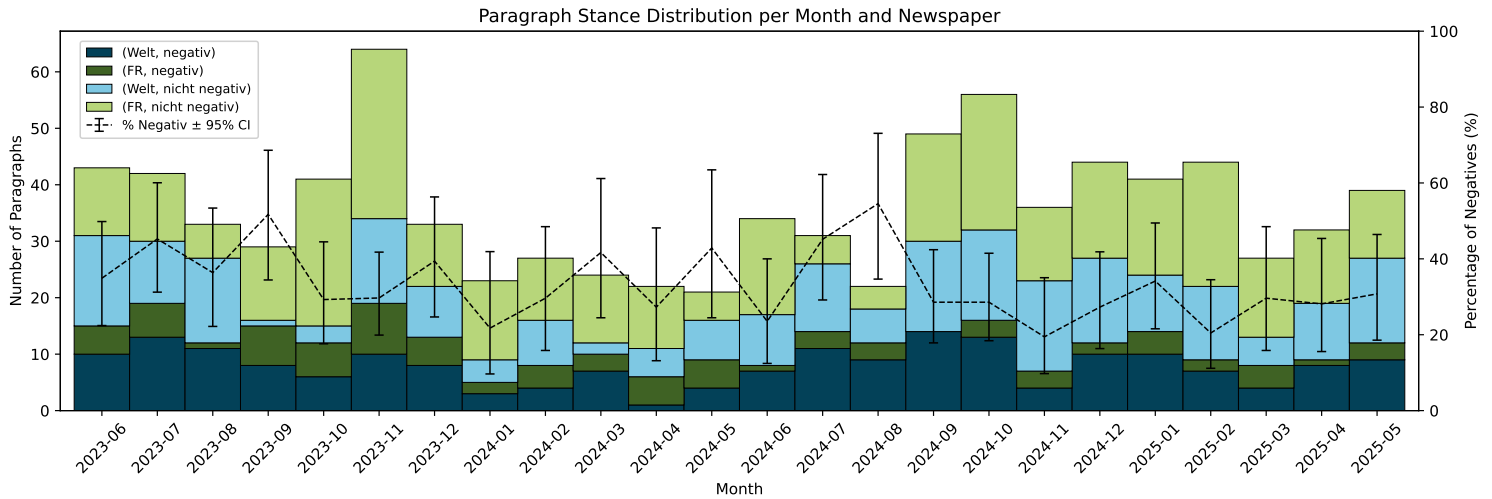


The implementation of the reform of the "Fachkräfteeinwanderungsgesetz" (FEG) with the goal of promoting the inflow of skilled workers from abroad went off in three steps between November 2023 and July 2024 (Bundesministerium für Wirtschaft und Energie, 2024). It is clearly visible in the plot that each of the three phases of implementation caused an increase in the newspaper article output about the topic, while declining slightly with each step. Over this period, *FR* constantly published more articles about the topic than *Die Welt*, but when it comes to the following peak in September and October 2024, the output of *Die Welt* exceeded the one of *FR* slightly. In these two months, several events happened that probably influenced the German discourse about migration and labor. On the one hand there was a new migration agreement with Kenya and Uzbekistan, however also previous migration agreements did not seem to have a very severe effect on the plot (e.g. Georgia: December 2023; Morocco: January 2024) (Hanewinkel, 2025). While after a closer look at the individual articles and paragraphs, it becomes obvious that their coverage certainly plays a role in the output of the newspapers. Another significant event during that period was the Islamist attack in Solingen on August 23, 2024, which continued to receive media coverage into September, as well as the attack on the Israeli consulate in Munich on September 5, 2024. As an aftermath, tightening asylum laws had been implemented as part of a so called "security package", limiting social welfare benefits for asylum seekers (Hanewinkel, 2025). While these events are obviously not related in to the development of the labor market, it seems eye-opening that the reporting about labor and migration increases sharply during that period. A closer examination of the articles of September and October 2024 reveals that approximately one third focused on new border controls and the tightening of asylum laws. These articles often frame migration not as a solution to the

skilled labor shortage, but rather as a factor exacerbating the problem. The last event that jumps into the eye are the general elections in February 2025. There is a peak in publications, especially from *FR* while *Die Welt* remains on a stable level.

A closer look at the stance distribution of the overall dataset, labeled by the NLI classifier, reveals extensive insights, especially about the named events of interest.

Figure 1: Paragraph stance distribution per Month and Newspaper



A distinctive and expected finding is that *Die Welt* consistently has a higher amount and proportion of negative paragraphs, with a mean value of 46% of published paragraphs labeled as negative. In contrast, *FR* shows a mean share of only 21.1% negative paragraphs. This difference can be simply explained by the conservative, towards right-winged-liberal characteristics of *Die Welt*.

One hypothesis was, that during key events that correlate with negative assertions towards migrants, the proportion of negative paragraphs would rise. A closer look at the key events, however, reveals that the share of negative articles did not significantly increase in response. Only during the general elections, a slight increase in negative stance can be ascertained, however it contradicts with the preceding outliers during the first half of 2024.

These prominent outliers, culminating in the peak negative share distribution across both newspapers in August 2024 are difficult to interpret, since they seemingly do not correlate with any considerable key event. Therefore, these outliers may be an effect of the general data situation. What they have in common is a relatively low amount of cases with roughly 30 paragraphs per month, leading to an increased statistical effect of single paragraphs and possible noise data, making the plot harder to interpret. Especially August 2024 shows, that the low amount of *FR* paragraphs

during that month³, fail to "counter" the proportionally many negatives by *Die Welt*, leading to the outlying data point. To illustrate these ambiguities, I included a 95% confidence interval (CI) to the plot, indicating the uncertainty of the values. What becomes evident however is the counterbalancing effect that *FR* paragraphs exert in contrast to those of *Die Welt*.

Where the data situation is better though, findings are also easier to interpret. In the consecutive months that include the above mentioned events between September 2024 and February 2025, it is visible, that although paragraph-frequency increased significantly, the total negative share remained at a relatively stable throughout this period. Only the general elections, as noted before, exhibit an increase in negative paragraphs, although it should be noted that additional data would be necessary to draw more definitive conclusions.

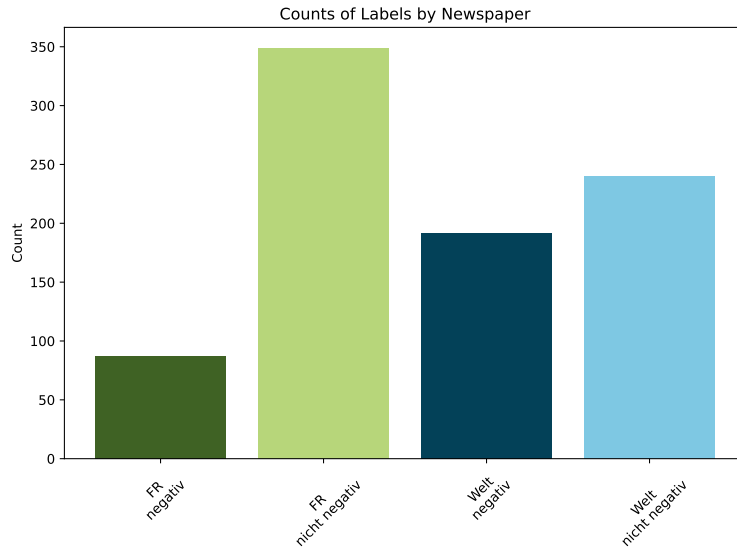
One initial idea of this research was to separate dataset A from dataset B to analyze stance development across different topics and discursive increments over time. The lack of data, however hampers this intend, since there would be just too few data points per month to draw reliable conclusions. What can be done however, is to map the over-all distribution of negative annotated paragraphs across both dataset groups. Paragraphs containing terms related to "Fachkräftemangel" and generally the lack of skilled workers (dataset B) are labeled more negative towards migration than general articles about migration and work related topics (dataset A). Both newspapers seem to be more negative about migration when it comes to "Fachkräftemangel". This seems counterintuitive, since researchers are in agreement about the necessity and the positive effect of migration on the labor market (Fuchs, Söhnlein and Weber, 2021; Hickmann *et al.*, 2021; Foroutan *et al.*, 2024; Kubis and Schneider, 2024; Kuhn *et al.*, 2025; Newbery, 2025). On the other hand however the two datasets are framed differently: With A mapping a broad depiction of the topic and B concentrating on the negative aspects of it, specifically targeting word endings like "-mangel", "-not" and "-knappheit". In total, *Die Welt* has more than twice as many negative paragraphs than *FR*, while the total amount of *negativ*-labels across the whole dataset is 32.1%.

Table 5: Distribution of "negativ"-labels across newspapers and groups in %

Newspaper Group	FR	Die Welt	Total %
A	17.6	41.8	28.3
B	23.6	46.9	36.7
Total %	20.0	44.4	32.1

3 One theory that would need further examination to affirm could be, that the attack in Solingen lead to a restraint of *FR* to write about migration and the shortage of skilled workers (a mainly positively framed topic by the newspaper). However it could also just be an effect of the so called "silly season" or "Sommerloch".

Figure 3: Stance distribution bar chart



7. Conclusion & Limitations

An initial question of this paper was, whether the presented computational methods for text data annotation were capable of yielding adequate results in stance detection. Automated stance detection in newspaper articles remains a highly complex and challenging task due to the nuanced and context-dependent nature of journalistic language. Length of newspaper articles, as well as the linguistic and investigation-related decisions that often constitute the final stance an article add to the complexity of the task. Since input size has a negative impact on the performance of large language models (LLMs), paragraph separation and selection can achieve a fair balance between contextualization and processability by computational algorithms. I demonstrated the applicability of pretrained large language models for this task, annotating German newspaper articles about migration and shortage of skilled workers. While this is a highly complex sociopolitical issue, it is comprehensible that the open models used had difficulties annotating three stance categories accurately. Providing additional context and instructions to the generative LLMs could not upgrade their performance significantly, but llama performed best with short and concise prompt instructions, while gemma profited from detailed descriptions of stance.

Across all models, especially the neutral class was difficult to detect properly, a problem acknowledge by researchers across the field (Mohammad, Sobhani and Kiritchenko, 2017, p. 2; Burnham, 2024, p. 5; Lüüsi *et al.*, 2024, p. 19; Lee *et al.*, 2025, p. 17). Currently, the language models used, seem not to be capable of performing a multi-class stance detection task on unseen

and complex textual data off the hand. However, reducing task complexity to a binary classification significantly improves model performance, suggesting that such an application could be useful for annotation tasks on real-world data. The usage of generative LLMs for stance detection is a valid method to adjust context knowledge and task specifications, although the open models presented still fall behind optimized NLI classifiers for stance detection in political texts. Considering resource efficiency NLI classifiers should be advised by now, but finetuning and further development of generative LLMs are likely scenarios to improve their capabilities in this specific task.

It must be acknowledged that selecting a less complex task would probably have contributed to a more substantial interpretability of the data. The restriction of newspaper articles to specific subtopics, namely migration and the shortage of skilled workers, led to an insufficient data situation, especially for examining the over-time development of stance distribution⁴. It is worth noting that an increased amount of cases per data point will lead to more statistical confidence and therefore better interpretability of the data. As a general rule, more data leads to clearer results, especially when plotting over time. The conducted analysis of the data should therefore be perceived as an exemplary proposition, demonstrating possibilities, trends and limitations of the data.

One clear result is that *Die Welt* reports significantly more negative than *FR* on migration and the shortage of skilled workers, which is noteworthy since its conservative, yet market-liberal ideology would allow theories in either way, considering that migration is an economic necessity to prevent recession in Germany. To draw more meaningful conclusions however, a multi-class analysis of stance would be necessary, since a binary classification remains inconclusive about the relation of negative and positive paragraphs published by *Die Welt*.

Follow-up research should further explore the possibilities of low-threshold binary classification tasks of NLI and in-context classifiers as long as non-finetuned classifiers are not sufficient enough to be applied directly to unseen data. Then, an extensive dataset would be suggested to clearly recognize trends and tendencies within the data. Another promising approach could involve machine-translating NLI premises prior to classification, as the most advanced classifiers currently operate exclusively in English, as Moritz Lauerer *et al.* (2023) suggest. As machine-translating technology improves steadily, this can be a way to extend research application to smaller languages without the need of finetuning and large annotated datasets. This could also lead to new insights,

4 However, not specifying the data and concentrate solely on migration would have lead to a massive increase of available data, leading to subsequent issues in downloading (limitation of LexisNexis downloads) and processing (limitation of RAM) of the data.

explaining why english-only models sometimes exceed performance of some multilingual models, as shown in this paper.

The python code and data of this paper is to be found at:

<https://github.com/hanauert/Hausarbeit-GenAI>

Large parts of the code were written with the assistance of ChatGPT (OpenAI), which was used for generating code, suggesting code structure, and debugging. All outputs were reviewed, tested, and adapted by the author to ensure correctness and relevance.

8. References

- ALDayel, A. and Magdy, W. (2021) ‘Stance detection on social media: State of the art and trends’, *Information Processing & Management*, 58(4), Available at: <https://doi.org/10.1016/j.ipm.2021.102597>.
- Arlt, D. and Wolling, J. (2018) ‘Bias wanted! Examining people’s information exposure, quality expectations and bias perceptions in the context of the refugees debate among different segments of the German population’, *Communications*, 43(1), pp. 75–99. Available at: <https://doi.org/10.1515/commun-2017-0045>.
- Bundesministerium für Wirtschaft und Energie (2024) *Das neue Fachkräfteeinwanderungsgesetz auf einen Blick*, *make-it-in-germany.com*. Available at: <https://www.make-it-in-germany.com/de/visum-aufenthalt/fachkraefteeinwanderungsgesetz> (Accessed: 13 August 2025).
- Burnham, M. (2024) ‘Stance detection: a practical guide to classifying political beliefs in text’, *Political Science Research and Methods*, 13(3), Available at: <https://doi.org/10.1017/psrm.2024.35>.
- Burnham, M., Kahn, K., Wang, R.Y. and Peng, R.X. (2024) ‘Political DEBATE: Efficient Zero-shot and Few-shot Classifiers for Political Text’. arXiv. Available at: <https://doi.org/10.48550/arXiv.2409.02078>.
- Cruickshank, I.J. and Ng, L.H.X. (2024) ‘Prompting and Fine-Tuning Open-Sourced Large Language Models for Stance Classification’. arXiv. Available at: <https://doi.org/10.48550/arXiv.2309.13734>.
- Darwish, K., Stefanov, P., Aupetit, M. and Nakov, P. (2020) ‘Unsupervised User Stance Detection on Twitter’, *Proceedings of the International AAAI Conference on Web and Social Media*, 14, pp. 141–152. Available at: <https://doi.org/10.1609/icwsm.v14i1.7286>.
- DIE ZEIT (2023) ‘Bundestag beschließt Reform des Fachkräfteeinwanderungsgesetzes’, *DIE ZEIT*, 23 June. Available at: <https://www.zeit.de/politik/deutschland/2023-06/fachkraeftemangel-bundestag-einwanderungsgesetz-reform> (Accessed: 1 August 2025).
- Diermeier, M., Bergmann, K., Zink, B. and Päßler, N. (2025) ‘Rechtsaußen-Erstarken in Deutschland: Implikationen für den Wirtschaftsstandort’, *Institut der deutschen Wirtschaft Köln e. V.* [Preprint].
- Eikmanns, F. and Sittenauer, A. (2025) ‘Sicherheit, aber menschlich’, *taz*, 20 February. Available at: <https://taz.de/Nach-Taten-in-Muenchen-und-Aschaffenburg/!6070749/> (Accessed: 1 August 2025).
- España-Bonet, C. (2023) ‘Multilingual Coarse Political Stance Classification of Media. The Editorial Line of a ChatGPT and Bard Newspaper’, in *Findings of the Association for Computational Linguistics: EMNLP 2023. Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore: Association for Computational Linguistics, pp. 11757–11777. Available at: <https://doi.org/10.18653/v1/2023.findings-emnlp.787>.

- Fendel, T. and Ivanov, B. (2024) ‘Fachkräfteeinwanderungsgesetz’, *Institut für Arbeitsmarkt- und Berufsforschung (IAB) der Bundesagentur für Arbeit*, IAB-Kurzbericht Nr. 21. Available at: <https://doi.org/10.48720/IAB.KB.2421>.
- Foroutan, N., Oltmer, J., Klie, A.W., Fröhlich, C., *et al.* (2024) ‘Flucht und Migration’, *Aus Politik Und Zeitgeschichte*, 45/2024.
- Fuchs, J., Söhnlein, D. and Weber, B. (2021) ‘Projektion des Erwerbspersonenpotenzials bis 2060 - Demografische Entwicklung lässt das Arbeitskräfteangebot stark schrumpfen’, *Institut für Arbeitsmarkt- und Berufsforschung (IAB) der Bundesagentur für Arbeit* [Preprint].
- Hanewinkel, V. (2025) *Migrationspolitik – September und Oktober 2024*, Bundeszentrale für politische Bildung. Available at: <https://www.bpb.de/themen/migration-integration/migrationspolitik/558202/migrationspolitik-september-und-oktober-2024/#node-content-title-2> (Accessed: 11 August 2025).
- Hickmann, H., Jansen, A., Pierenkämper, S. and Werner, D. (2021) *Ohne sie geht nichts mehr: Welchen Beitrag leisten Migrant_innen und Geflüchtete zur Sicherung der Arbeitskräftebedarfe in Fachkraftberufen in Deutschland?* Bonn: Friedrich-Ebert-Stiftung (FES diskurs).
- InterNations (2023) *Expat Insider 2023*. Available at: <https://www.internations.org/content-assets/static/3ec4c33380c9b0be17901f29117421da/ei2023.pdf> (Accessed: 19 August 2025).
- Kubis, A. and Schneider, L. (2024) ‘Zuwanderung und Arbeitsmarkt: Eine Analyse für Deutschland und die Bundesländer’. Available at: <https://doi.org/10.11586/2024166>.
- Kuhn, S., Schwengler, B., Seibert, H. and Wiethölter, D. (2025) ‘Demografischer Wandel und Arbeitskräftemangel: Ausländische Beschäftigte spielen eine wichtige Rolle auf dem Arbeitsmarkt’, *Institut für Arbeitsmarkt- und Berufsforschung (IAB) der Bundesagentur für Arbeit*, IAB-Kurzbericht Nr. 8. Available at: <https://doi.org/10.48720/IAB.KB.2508>.
- Laurer, M., Atteveldt, W. van, Casas, A. and Welbers, K. (2024) ‘Building Efficient Universal Classifiers with Natural Language Inference’. arXiv. Available at: <https://doi.org/10.48550/arXiv.2312.17543>.
- Lee, D., Choi, J., Han, J. and Park, K. (2025) ‘Journalism-Guided Agentic In-Context Learning for News Stance Detection’. arXiv. Available at: <https://doi.org/10.48550/arXiv.2507.11049>.
- Loschert, F., Leisenheimer, M. and Komitowski, D. (2025) *Willkommen zurück? Abwanderungsgründe und Rückwanderungspotenziale von aus Deutschland abgewanderten EU- und Drittstaatsangehörigen*. Berlin: Friedrich-Ebert-Stiftung (Analyse / Friedrich-Ebert-Stiftung).
- Lüüsi, L., Kangur, U., Chakraborty, R. and Sharma, R. (2024) ‘Political Stance Detection in Estonian News Media’, in M. Härmäläinen, F. Pirinen, M. Macias, and M. Crespo Avila (eds) *Proceedings of the 9th International Workshop on Computational Linguistics for Uralic Languages*. Helsinki, Finland: Association for Computational Linguistics, pp. 12–28. Available at: <https://aclanthology.org/2024.iwclul-1.2/>.
- Mascarell, L., Ruzsics, T., Schneebeil, C., Schlattner, P., *et al.* (2021) ‘Stance Detection in German News Articles’, in *Proceedings of the Fourth Workshop on Fact Extraction and VERification*

(FEVER). Association for Computational Linguistics, pp. 66–77. Available at: <https://doi.org/10.18653/v1/2021.fever-1.8>.

Mets, M., Karjus, A., Ibrus, I. and Schich, M. (2024) ‘Automated stance detection in complex topics and small languages: The challenging case of immigration in polarizing news media’, *PLOS ONE*. Edited by N. Grabar, 19(4), Available at: <https://doi.org/10.1371/journal.pone.0302380>.

Mohammad, S.M., Sobhani, P. and Kiritchenko, S. (2017) ‘Stance and Sentiment in Tweets’, *ACM Transactions on Internet Technology*, 17(3), pp. 1–23. Available at: <https://doi.org/10.1145/3003433>.

Moritz Laurer, Wouter Van Atteveldt, Andreu Casas, and Kasper Welbers (2023) ‘Lowering the Language Barrier: Investigating Deep Transfer Learning and Machine Translation for Multilingual Analyses of Political Texts’, *Computational Communication Research*, 5(2), Available at: <https://doi.org/10.5117/CCR2023.2.7.LAUR>.

Newbery, H. (2025) *Fachkräftesicherung und Zuwanderung: Bilanz, Hürden und Perspektiven*. Bonn: Friedrich-Ebert-Stiftung.

Riedl, A.A. (2024) *Nachrichtenqualität als journalistischer Prozess*. DE: Herbert von Halem Verlag. Available at: https://doi.org/10.1453/OA_2024_9783869626611 (Accessed: 31 July 2025).

Schetsche, M. (2001) ‘Metatheorie und Analyseprogramm: zum Doppelstatus der relativistischen Problemtheorie’, *Soziale Probleme*(12(1/2)), pp. 28–44. Available at: <https://nbn-resolving.org/urn:nbn:de:0168-ssoar-248352>.

Appendix A

1.1 LexisNexis keyword query:

Dataset A	Dataset B
01.06.2023 - 15.06.2025	01.06.2023 - 15.06.2025
(Arbeit! OR Job! OR Personal!) AND (Migrat! OR Migrant!)) AND NOT (Fachkräftemangel! OR Fachkräfte! OR Fachkraft!)	(Fachkräftemangel! OR Fachkräfte!) AND (Migrat! OR Migrant!)

1.2 Paragraph splitting

Regex dataset A	Regex Dataset B
r'(?=.*\b(?:Migrat\w* Migrant\w* Zuwanderung\w* Einwanderung\w* Geflüchtet\w* Flüchtling\w* Drittstaat\w*)\b)'	r'(?=.*\b(?:Migrat\w* Migrant\w* Zuwanderung\w* Einwanderung\w* Geflüchtet\w* Flüchtling\w* Drittstaat\w*)\b)'
r'(?i)\b(?:Jobangebote? Arbeitsplätz(?:e en)? Arbeitsmarkt\w* Beruf\w* Anstellung\w* Erwerbstätig\w* Lohn\w* Gehalt\w* Einkommen\w* Ausbildung\w* Qualifikationen? Beschäftigung(?:sverhältnis smöglichkeiten schancen)? Vollbeschäftigung Teilzeitbeschäftigung Beschäftigten)\b)'	r'(?=.*\b(?:Fachkräft\w* Fachkräf\w* Arbeitskräfte(?:mangel engpass not problem knappheit)? Personal(?:mangel engpass not problem knappheit)? qualifizierte\w*\s+Arbeitskräfte?)\b)'

1.3

Dataset Stats

Dataset A

newspaper	unique_articles	paragraphs
FR	218	262
Welt	144	208
Total	362	470

Dataset B

newspaper	unique_articles	paragraphs
FR	129	174
Welt	142	224
Total	271	398

Total share

newspaper	unique_articles (%)	paragraphs (%)
FR	54.82	50.23
Welt	45.18	49.77
Total	100.00	100.00

Total word count FR & Welt

word count	max	min	mean
paragraph	371.0	5.0	103.66
body_text	2489.0	67.0	835.87

Appendix B

Annotation guidelines to determine whether a paragraph is positive, negative or neutral towards immigration and migrants were the following:

- *Der Text spricht eher positiv/negativ/neutral über Migration oder Migrant*innen.*

If unclear, decide according to follow-up questions:

- *Reproduziert der Artikel eine migrationsfeindliche Stimmung?*
- *Sieht der Artikel Migration eher als Problem, als Lösung oder irrelevant für sozioökonomische Probleme?*

Definition:

- *positiv: Der Text zeigt eine befürwortende Haltung gegenüber Migration.*
- *negativ: Migration wird im Text eher kritisch betrachtet.*
- *neutral/irrelevant: Der Text drückt keine eindeutige Meinung zu Migration aus. Der Text hat keinen Bezug zum Thema.*

Borderline cases:

"Migration wird es immer geben. Das ist ein Fakt. Die Frage ist, wie wir damit umgehen. Und da sagen wir: Geflüchtete Menschen müssen schnell die Möglichkeit bekommen, zu arbeiten. Arbeit sorgt dafür, dass Akzeptanz da ist. Jeden Tag redet irgendeine Branche von Arbeitskräfte- und Fachkräftemangel. Gleichzeitig gibt es Menschen, die teilweise schon länger in Sachsen leben und nicht arbeiten, weil sie nicht dürfen. Wir müssen deutlich machen, dass die Arbeitskraft der Geflüchteten hier gewollt ist. Auch die Arbeitgeber haben erkannt, dass es dort ein Potenzial gibt, das wir zurzeit noch nicht nutzen."

*Neutral. Migrant*innen werden zwar als auch als erwünscht bezeichnet, jedoch vollständig auf ihre Nützlichkeit und Arbeitskräfte reduziert. Technokratisch, funktional.*

"Was erhoffen sich die jungen Wähler von der AfD?"

Maas: Eine Verbesserung der Lage, wobei die Richtung diffus blieb. Viele erhoffen sich, dass dann mehr Geld an Bedürftige geht. Dass die alte Oma keine Flaschen mehr sammeln muss. Die Flüchtlinge werden als Schmarotzer dargestellt, die alles geschenkt bekommen. Das ist derselbe Futterneid, mit dem früher gesagt wurde: Die Ausländer nehmen uns die Arbeitsplätze weg. Jetzt heißt es: Die arbeiten zu wenig. Es herrscht offenbar das Gefühl vor: Mir geht es zwar gut, aber es könnte mir auch noch viel besser gehen, wenn die Flüchtlinge uns nicht so viel Geld kosten würden."

Negativ. Obwohl der Sprecher anderer Meinung ist reproduziert er rassistische Stereotype der AfD. AfD-Versteher.

"Die Bundesregierung hat mit Usbekistan ein Migrationsabkommen ausgehandelt, das auf der anstehenden Reise von Kanzler Olaf Scholz nach Zentralasien unterzeichnet werden soll. Das erfuhr

Reuters aus Regierungskreisen. Ziel des Abkommens sei vor allem die Ausbildung und Anwerbung von Arbeitskräften aus Usbekistan. Die Rücknahme abgelehnter Asylbewerber spiele in diesem bilateralen Abkommen eine geringere Rolle. Dem Vernehmen nach gibt es keine Verbindung zwischen diesem seit Wochen ausgehandelten Abkommen und einer möglichen Abschiebung afghanischer Straftäter. Scholz reist am Sonntag nach Usbekistan und Kasachstan, wo er mit Regierungsspitzen aller fünf zentralasiatischen Länder zusammentreffen wird."

Negativ, da impliziert wird, dass eigentlich Abschiebungen nötig sind. Latente Kritik an dem Abkommen, da "seit Wochen" verhandelt würde, jedoch Abschiebungen nur eine "geringere" Rolle spielen würden.

"Ihre Mordserie ist in Deutschland beispiellos. Neun Männer sterben. Hingerichtet an ihren Arbeitsplätzen. Ein Blumenstand. Ein Imbiss. Oder eine Schneiderei. Erschossen am helllichten Tag. Weil sie Migranten sind. Das Trio ermordet auch eine Polizistin."

Kontextwissen NSU. Off-topic. Neutral.

"Der aus dem Irak kommende Flüchtling hat eine Ausbildung als Koch angetreten. Von dem Fachkräftemangel im Hotel- und Gaststättengewerbe scheint von den Verantwortlichen für diese Abschiebungen niemand gehört zu haben."

Positiv, da die Ausbildung zum Koch als positiv beurteilt wird. Hier noch zusätzlich erschwerend: Ironie.

"Früher, sagt Mohamed Ibrahim, „habe ich mich gerne als Deutsch-Palästinenser vorgestellt. Jetzt lasse ich das Deutsch lieber weg.“ Dabei ist der 54-jährige Familienvater, sportliche Figur, stoppelkurzes Haar, ein Musterexemplar gelungener Integration. Einer, der als Flüchtlingsjunge unter widrigen Bedingungen seinen Weg gemacht hat. Und der sich heute neben seinem Beruf als Berater für Entwicklungskooperation ein- bis zweimal pro Monat in Workshops über Wissensvermittlung und Perspektivwechsel im Nahostkonflikt engagiert."

Neutral. Migration wird nicht per se als gut oder schlecht dargestellt. Negative Migrationserfahrungen stehen hier gegen "gelungene Integration".

"Im Jahr 1975 sind Sie aus der heutigen Demokratischen Republik Kongo nach Belgien gekommen. Erst sieben Jahre später wurden Sie als politischer Flüchtling anerkannt. Wie blicken Sie auf die Zeit zurück? Kompany: Ich hatte eine wunderbare Kindheit. Mein Vater und meine Mutter kamen aus wohlhabenden Verhältnissen. In unserer Familie gab es keine finanziellen Probleme, und wir konnten alle eine gute Ausbildung genießen. Ich wollte in meiner Heimat bleiben, nur musste ich irgendwann gehen, weil ich mich dort gegen die Diktatur gestellt habe dann wurde es hart. Ich lebte sieben Jahre in Belgien ohne offizielle Dokumente. Das bedeutet, dass ich keine Rechte hatte keine offizielle Arbeit, keine feste Perspektive. Ich war quasi ein Niemand. Ich musste mich irgendwie durchschlagen, aber ich fand meinen Weg. Ich spielte Fußball, begann, junge Spieler zu trainieren. Ich wusste, dass ich mich in die Gesellschaft einbringen musste, wenn ich akzeptiert werden wollte. In Belgien habe ich schnell verstanden, wie ich zwischen den Menschen leben muss. Ich habe mich nicht provozieren lassen, ich bin friedlich geblieben, aber immer sehr aufmerksam."

Positiv, da eine Erfolgsgeschichte erzählt wird. Dennoch werden vor allem negative Migrationserfahrungen benannt.

Appendix C

1.1 Multi-class Prompts for NLI Classifiers:

1. Prompt:

```
template = 'Der Autor diese Artikels vertritt eine {} zu Migration.'  
labels = ['positive', 'negative', 'neutrale']
```

2. Prompt:

```
template = 'Der Artikel positioniert sich {} zu Migration.'  
labels = ['positiv', 'negativ', 'neutral']
```

3. Prompt:

```
template = 'Der Artikel vertritt die Meinung, dass sich Migration {} auf  
den Fachkräftemangel auswirkt.'  
labels = ['positiv', 'negativ', 'neutral']
```

4. Prompt (proposed by Chat-GPT)

```
template = "Migration wirkt sich {} auf den Fachkräftemangel aus."  
labels = ['positiv', 'negativ', 'neutral']
```

5. Prompt

```
template = "Der Autor diese Artikels vertritt eine {} Meinung zu  
Migration in Bezug auf Fachkräftemangel."  
labels = ['positive', 'negative', 'keine']
```

6. Prompt

```
template = "Der Text impliziert eine {} Meinung zu Migration in Bezug auf  
Fachkräftemangel."  
labels = ['positive', 'negative', 'keine']
```

7. Prompt

```
template = 'Die Haltung des Autors zur Wirkung von Migration auf den  
Fachkräftemangel ist {}.'  
labels = ['positiv', 'neutral', 'negativ']
```

8. Prompt

```
template = 'Der Text positioniert sich {} zu Migration.'  
labels = ['positiv', 'negativ', 'neutral']
```

Several tests have shown that best results are achieved with applying the classifier to a combination of title and paragraph of the article, separated by linebreaks and with a short description of each part:

```
samples = list("Title:\n" + subsample_gold['title'] + "\n\n" + "Paragraph:\n" +  
subsample_gold['paragraph'])
```

NLI Classifiers:

- MoritzLaurer/bge-m3-zeroshot-v2.0
- mlburnham/Political_DEBATE_DeBERTa_large_v1.1
- luissattelmayer/NLI-stance-immigration-burnham-finetuned-german

1.2 Multi-class prompts for generative LLMs:

1. Prompt:

```
messages = [
```

```

    {"role": "system", "content": "Bitte hilf mir, die Haltung des folgenden Absatzes zu Migration zu klassifizieren."},
    {"role": "user", "content": f"{text} --- Wie ist die Haltung des Autors zu Migration?"}
]

```

2. Prompt:

```

messages = [
    {
        "role": "system",
        "content": (
            "Stance Detection: Klassifiziere den folgenden Absatz.\n"
            "positiv: Der Text zeigt eine befürwortende Haltung gegenüber Migration.\n"
            "negativ: Migration wird im Text eher kritisch betrachtet.\n"
            "neutral: Der Text drückt keine eindeutige Meinung zu Migration aus."
        )
    },
    {
        "role": "user",
        "content": f"{text} --- Wie ist die Haltung des Autors zu Migration? "
    }
]

```

3. Prompt:

```

messages = [
    {
        "role": "system",
        "content": (
            "Du bist ein hilfreicher Assistent für Stance Detection.\n"
            "Analysiere den folgenden Zeitungsabschnitt und klassifiziere die Haltung des Autors gegenüber dem Thema Migration.\n"
            "Wähle eine der folgenden Kategorien:\n"
            "- **positiv**: Der Autor unterstützt Migration oder spricht positiv darüber.\n"
            "- **negativ**: Der Autor lehnt Migration ab oder äußert sich kritisch.\n"
            "- **neutral**: Der Text zeigt keine klare Meinung oder bleibt ausgewogen.\n"
        )
    },
    {
        "role": "user",
        "content": f"{text}"
    }
]

```

The tests showed that shorter, less complex prompts lead generally to better results. While for gemma a combination of title and paragraph was best, llama profited from adding the lead as context information.

1.3 Binary classification prompts:

NLI Prompt:

```
template = 'Migrantinnen und Migranten werden in dem Text eher {} dargestellt.'  
labels = ['als Problem', 'nicht als Problem']
```

Gemma3:4b Prompt:

```
messages = [  
    {  
        "role": "system",  
        "content": (  
            "Stance Detection: Klassifiziere den folgenden Absatz.\n"  
            "negativ: Migration wird im Text eher kritisch betrachtet.\n"  
            "nicht negativ: Migration wird in dem Text positiv oder neutral  
betrachtet.\n"  
        )  
    },  
    {  
        "role": "user",  
        "content": f"{text} --- Wie ist die Haltung des Autors zu Migration? "  
    }  
]
```